CSE558 · DATA SCIENCE — LECTURE 5

Hypothesis Testing: From Suspicion to Decision

Your friend says she can read the cards. Today we build the machinery that decides.

Format

Continue experiment, peer instruction, and prediction rounds

Builds on

Lecture 4 — random variables and the CLT

Leads to

Lecture 6 — Test for independence, variance, groups etc

Supratim · IIIT-Delhi · Monsoon 2026

WHERE WE LEFT OFF

Lecture 4, in four moves

Binomial

n independent trials, each a success or a failure. Counts how many landed.

Geometric

How long until the first success? E[X] = 1/p.

Coupon Collector

Linearity of expectation turns one hard question into k easy ones.

Central Limit Theorem

Sum enough independent draws and the total goes normal — whatever you started from.

… and one question we deliberately walked away from.

Toy experiment

1. Shuffle, and test
2. Twenty draws, twenty guesses
3. 8 correct out of 20
4. $\overline{p} =$ 0 4

COMMIT BEFORE YOU LOOK

How many would have convinced you?

Guessing at random gives 5 correct out of 20. What is the smallest number of correct calls that would have convinced you that there were no guessing.

7 or 8

10

12 or 13

15+

THE UNFINISHED BUSINESS

Can your friend really read the cards?

She claims she can name the suit of a randomly drawn card more often than one time in four.

Round 1

200 cards drawn.
She calls 53 correctly.

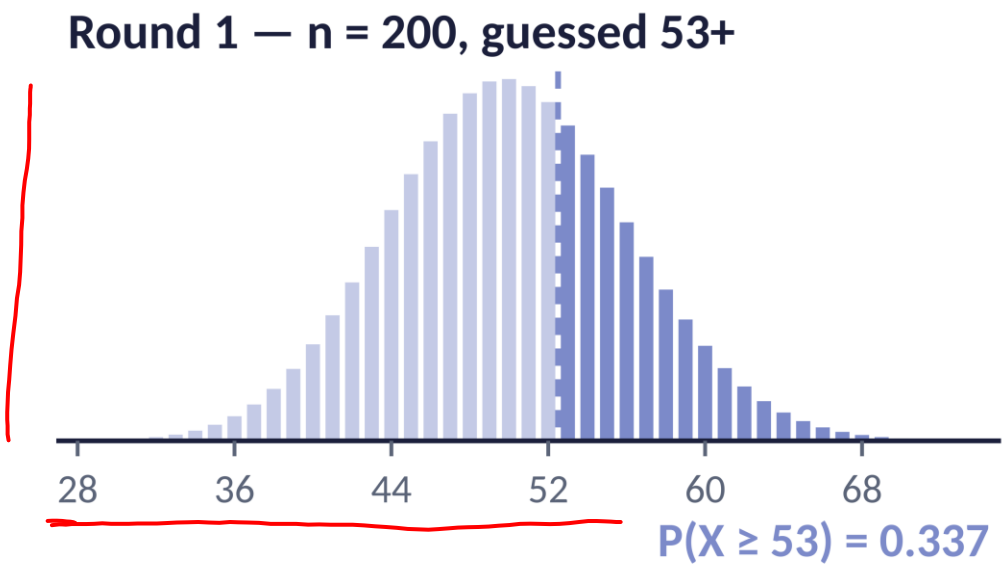
Round 2

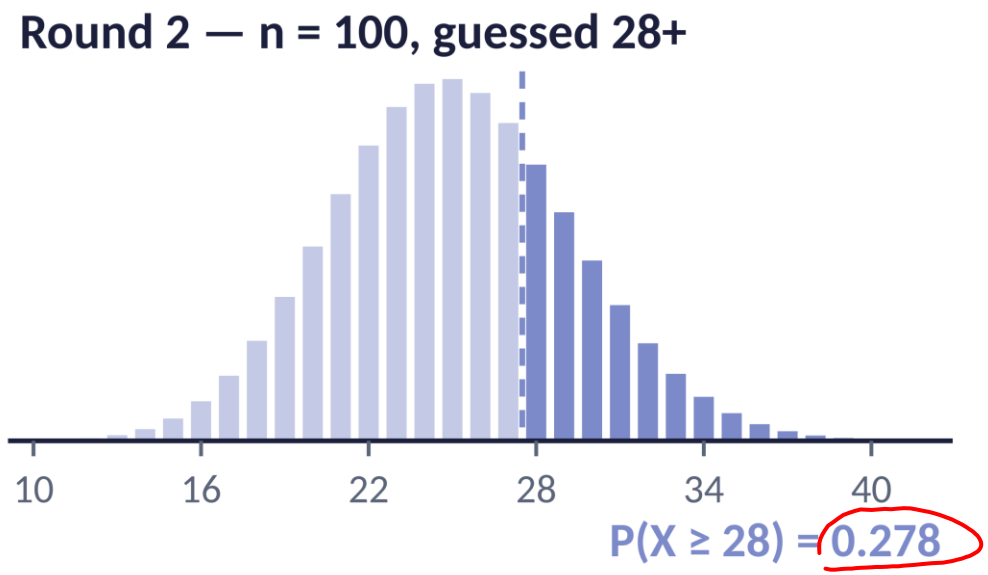
100 cards drawn.
She calls 28 correctly.

Round 3

100 cards drawn.
She calls 43 correctly.

If she is guessing, how often does this happen?



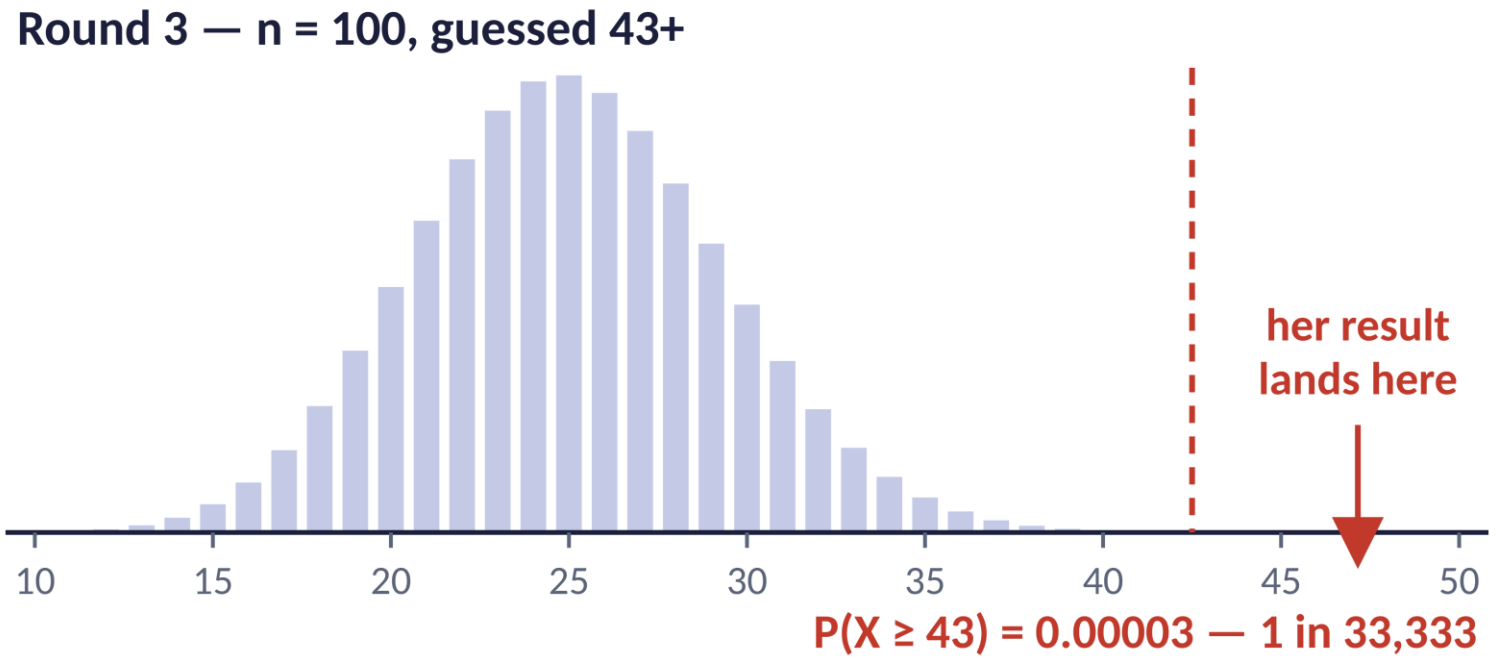


Both results sit comfortably inside what pure guessing produces.

A result you would see one time in three/four is not evidence of anything.

THEN THE THIRD ROUND

43 out of 100 changes everything



Same method. New data.

Nothing about the reasoning changed between round 2 and round 3. Only the number she hit did.

NAME THE MOVE

You have just run a hypothesis test

1. **Assume the trivial explanation**
   She is guessing. Every card is a one-in-four shot.

2. **Ask how likely the data is under it**
   $P(X \geq 43)$ when $n = 100$ and $p = 0.25$.

3. **Compare against a line you drew first**
   Below 5%? Below 0.01%? You decide before you look.

4. **Reject or do not reject the trivial explanation**
   You never prove she is psychic. You only rule luck out.

BEFORE YOU CAN TEST, YOU MUST ESTIMATE

What lives around a sample estimate?

The survey

"Medical science in the West is superior to India's."

31% of your sample agrees.

Standard error of that sample ≈ 0.01.

Empirical mean	0.31 — straight from the sample
Confidence level	95% ⇒ α = 0.05 ⇒ z = 1.96
Margin of error	1.96 × 0.01 = 0.0196
Confidence interval	0.31 ± 0.0196 = (0.29, 0.33)

Standard error

$$\sqrt{\frac{p(1-p)}{n}}$$

Why n is always small

Crash tests.
Adverse effects of a drug.

Rule of thumb

At least 15 successes and 15 failures before the normal approximation is safe.

THE VOCABULARY

Six terms, and you can read any test

H_0 — null hypothesis

The status quo. Treated as true until the data forces you off it.

H_1 — alternate hypothesis

The claim you are actually investigating.

T — test statistic

Distance between your estimate and the null value.

p-value

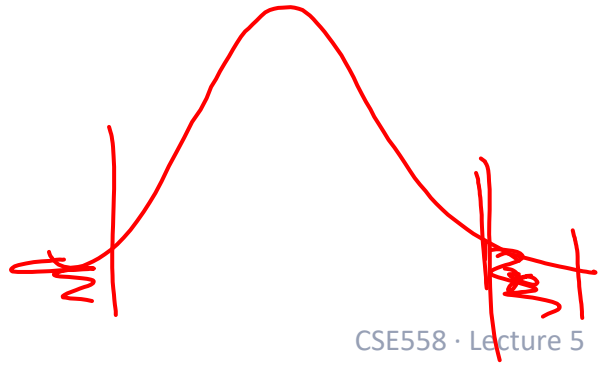
How likely data at least this extreme is, if H_0 holds.

Critical region

The values of T that would make you reject. One-sided or two.

α — level of significance

Your tolerance for a false alarm, fixed before you look.



A screening programme lowers its threshold

A cancer screening service decides to flag more people for follow-up. H_0: this patient is healthy. What happens to the error rates?

A α falls

B α rises

C Neither changes

Vote

Find a disagreement

Two minutes. Convince someone who voted differently.

Vote again

Two ways to be wrong

H_0: the defendant did not commit the crime. **H_1: the defendant committed the crime.**

Decision / Reality	H_0 is true: innocent	H_0 is false: guilty
Do not reject H_0: acquit	Correct decision	**Type II — β the guilty walk free**
Reject H_0: convict	**Type I — α an innocent is jailed**	Correct decision

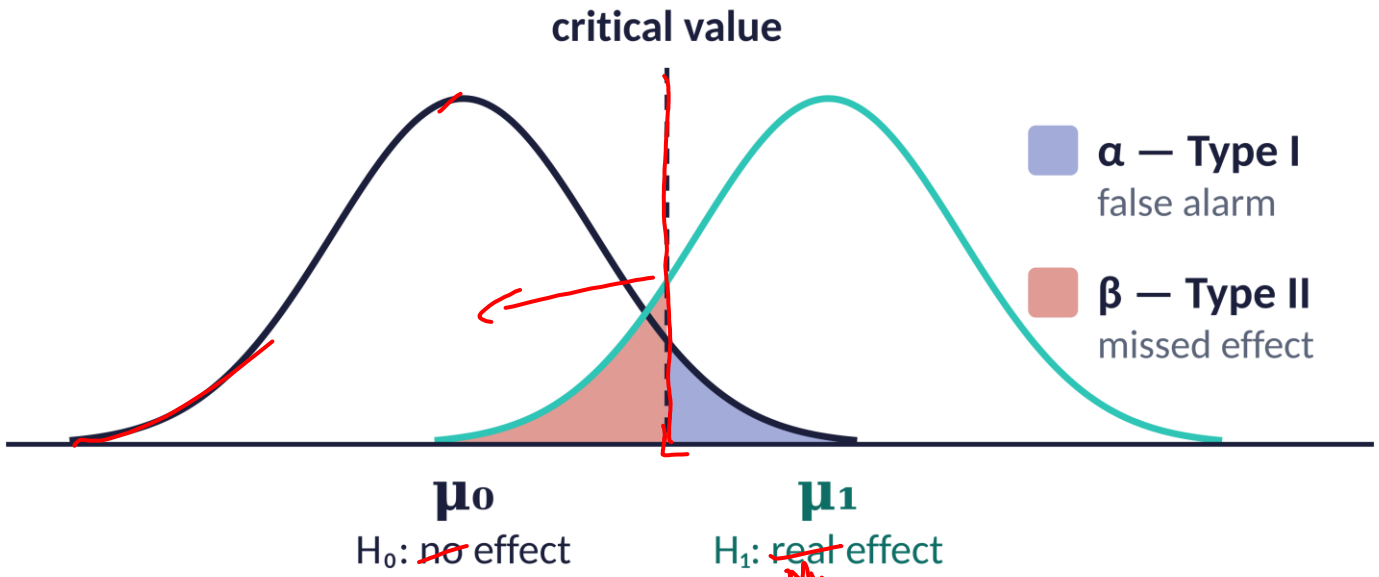
NEVER ACCEPT H_0

You reject it, or you fail to reject it.

A verdict of “not guilty” is not a finding of innocence.

THE TRADE-OFF YOU CANNOT ESCAPE

Does wearing a mask in NCR improve lung function?



Slide the line right and α shrinks — but β grows. You cannot shrink both.

α - level of sig
1 - α → confidence level

Set up the test

H_0: lung function is unchanged, mask or no mask.

H_1: lung function improves with a mask.

α is the risk of claiming a benefit that is not there.
β is the risk of missing one that is.

If you demanded 100% confidence, what would happen?

THE FIRST TEST

z-test for a population mean

Use it when

- One variable — a sample mean against a population mean.
- The population standard deviation σ is known.
- H_0: $\mu = \mu_0$
- H_1: $\mu \neq \mu_0$
- $\alpha = 0.05$

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

SE

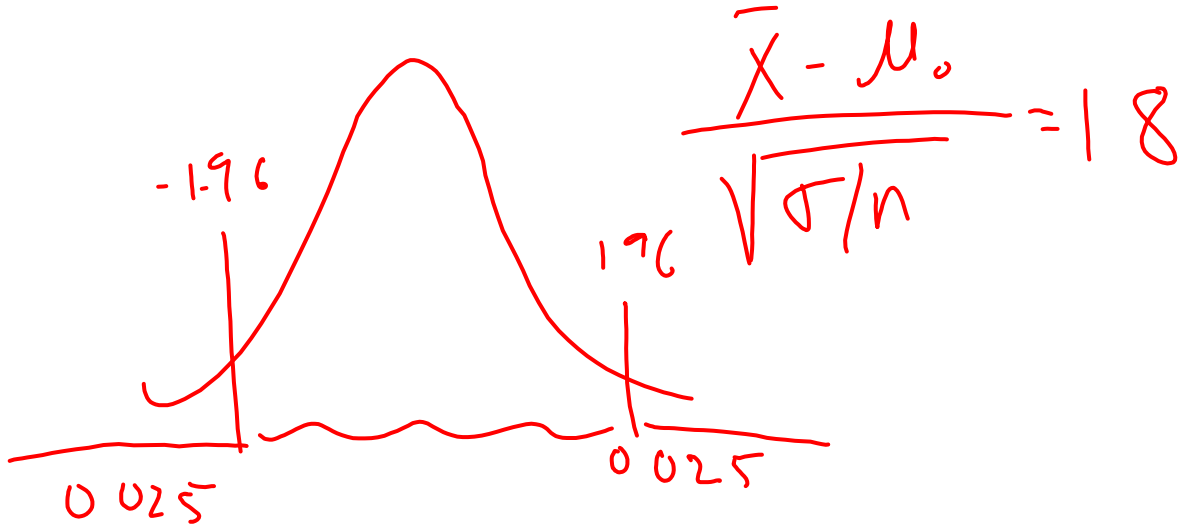
WORK IT THROUGH

$\mu_0 = 4.0,\ \sigma^2 = 1,\ n = 9,\ \bar{x} = 4.6$

Test at $\alpha = 0.05$, two-sided: $H_0: \mu = \mu_0$ against $H_1: \mu \neq \mu_0$

Z = 1.8

1.8 < 1.96, so do not reject H_0.



PREDICT

Same Z. Now change the question.

What you already know

Two-sided, H_1: $\mu \neq \mu_0$.
Critical value 1.96.
Z = 1.8 → do not reject.

Now suppose H_1: $\mu > \mu_0$

Identical data. Identical Z = 1.8.
Critical value 1.645
What is the verdict?

A Still do not reject

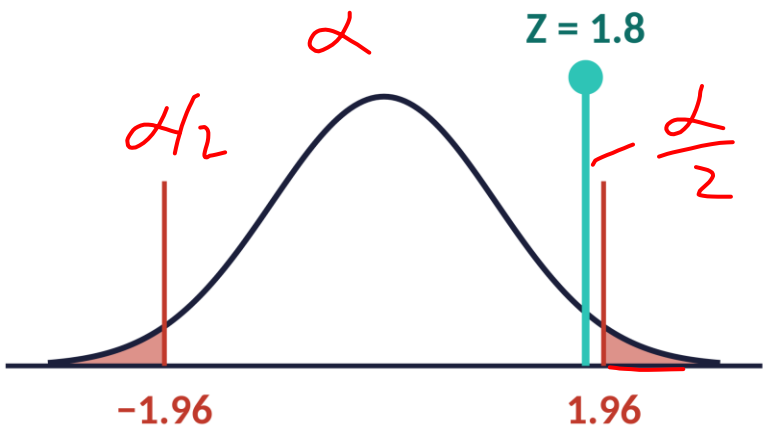
B Reject

C Cannot tell without more data

SAME NUMBERS, DIFFERENT QUESTION

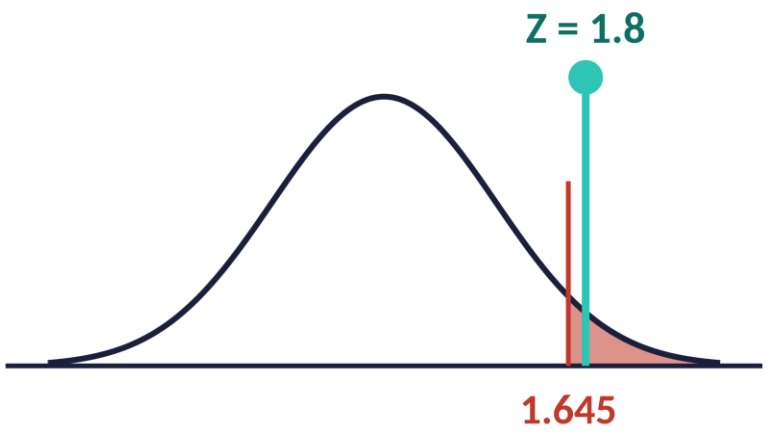
Change H_1 and the verdict flips

Two-sided — $H_1: \mu \neq \mu_0$



1.8 < 1.96 → do not reject

One-sided — $H_1: \mu > \mu_0$



1.8 > 1.645 → reject

Decide the direction before you see the data

BACK TO OUR OWN EXPERIMENT

Now settle it with the data we collected

Toy Experiment

H_o: $\mu \leq 0.25$ — guessing.
H_1: $\mu > 0.25$ — not guessing.
$\alpha = 0.05$, one-sided.
Critical Region: 1.645
n = 20 correct = 8

The given setup

Same H_o and H_1, same α.

n = 100, and she called 42 correctly, so $\bar{x} = 0.42$.

$$\frac{0.4 - 0.25}{\sqrt{\frac{0.25 \cdot 0.75}{20}}} =$$

YOUR TURN

The chip manufacturer

Q1 A fab claims at least 90% of its chips pass post-production testing. Your random sample has mean 89.98% and yields z = 1.65, with α = 0.05. Reject the claim, or fail to reject it?

Reject

Say, it is 1.65 against a critical value of 1.645.

The rule says reject!!

But is a fab delivering 89.98% instead of 90% actually a problem?

KNOW WHEN NOT TO USE IT

Conditions on the z-test

σ is known

You need the population standard deviation, not the sample's.

n is large

Roughly 30 or more observations.

Data are near normal

Otherwise the test approximates rather than answers.

Break the first one — σ unknown, which is almost every real dataset.

BEFORE YOU GO

Next lecture: First class test